

CrescendoShield: Multi-Layer Defense Against Multi-Turn Jailbreaks

Abstract

CrescendoShield is a reproducible AI safety prototype for detecting simulated Crescendo-style jailbreak attacks.

It integrates user input handling, risk detection, escalation analysis, memory sanitization, an LLM wrapper, and output verification.

Objective

Build a CPU-friendly research prototype that can detect, prevent, and report multi-turn jailbreak risk.

The project is designed for internship evaluation, classroom review, and GitHub reproducibility.

Pipeline

user_input -> risk_detector -> escalation_analyzer -> memory_sanitizer -> llm_wrapper -> output_verifier -> final_response

Dataset and Methodology

The benchmark contains 2,500 simulated text-only records across five categories.

Each category has 500 examples generated deterministically by `evaluation/benchmark.py`.

Categories

Benign: ordinary safe requests for false-positive testing.

Jailbreak: roleplay and instruction-bypass attempts.

Prompt injection: synthetic system/developer override patterns.

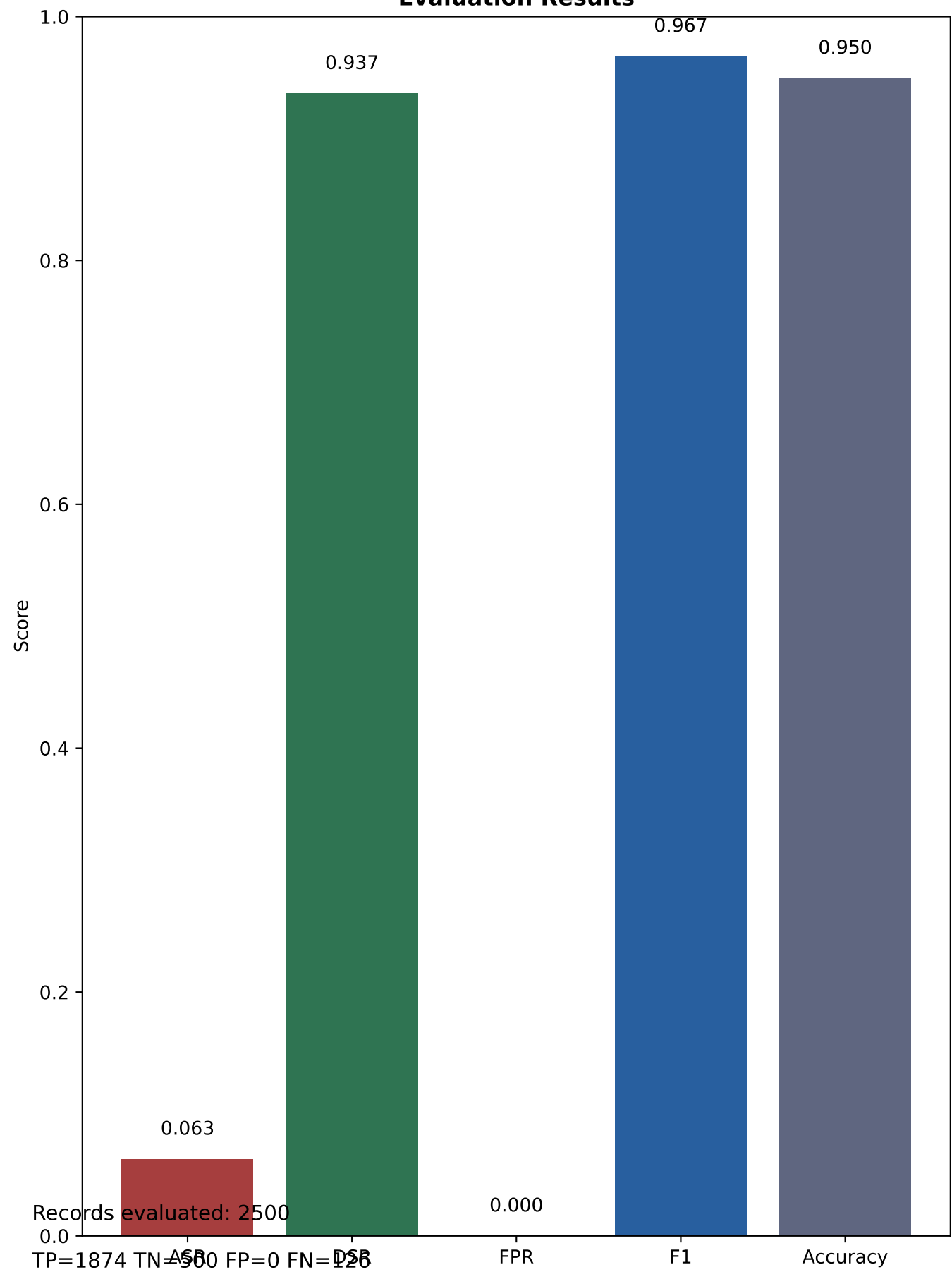
Escalation: multi-turn requests that move from harmless to risky.

Context poisoning: false authority, fake approval, and memory manipulation.

Safety Scope

The dataset avoids operational malware, phishing, chemical synthesis, and real harm instructions.

Evaluation Results



Limitations and Future Work

Limitations

The benchmark is simulated and pattern-centered, so it cannot replace human red-team evaluation.

The default LLM wrapper uses an offline-safe deterministic backend for reproducibility.

Real Llama-3.2-3B-Instruct evaluation requires model access, compute, and safety review.

Future Work

Add semantic embedding classifiers, multilingual examples, human-reviewed labels, and open-source guard model baselines.

Extend the dashboard with risk trajectory charts and model comparison experiments.

Conclusion

CrescendoShield provides a complete, reproducible, deployment-ready AI safety submission with code, data, metrics, dashboard, and report.